How do model processes affect HBV simulation?

Calibration and process-switching experiments

MMUQ Assignment 1 — Group presentation

Research question and presentation roadmap

Question: How does a calibrated HBV simulation respond when selected water pathways are switched off?

1. Calibrate a baseline against observed discharge.
2. Disable three groups of processes, keeping other parameters fixed.
3. Compare hydrographs, storage and performance metrics.
4. Discuss interpretation and limitations.

The focus is on **what the model does** when a process is removed.

Calibration: fitting HBV to the observed hydrograph

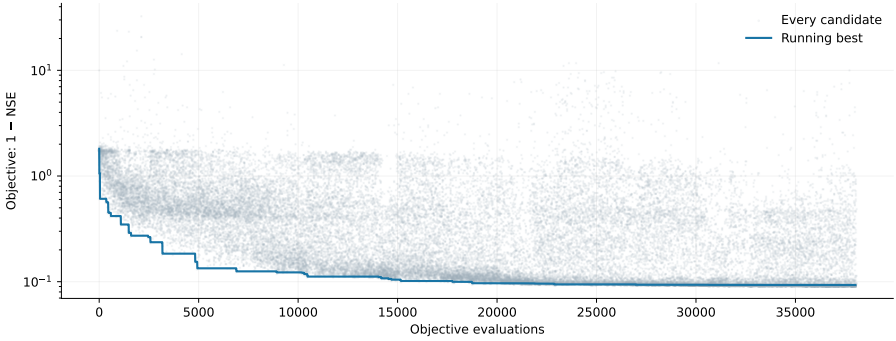
- Supplied HBV001a and hourly inputs; catchment area 959 km^2.
- Nov 2010–Feb 2011: 2,880 hourly steps.
- 18 calibrated entries, including four initial storages.
- Differential Evolution (DE) searches the supplied parameter bounds.

Objective: minimise

$$J(\theta) = 1 - \mathrm{NSE} = \frac{\sum_t (Q_{sim,t} - Q_{obs,t})^2}{\sum_t (Q_{obs,t} - \overline{Q}_{obs})^2}.$$

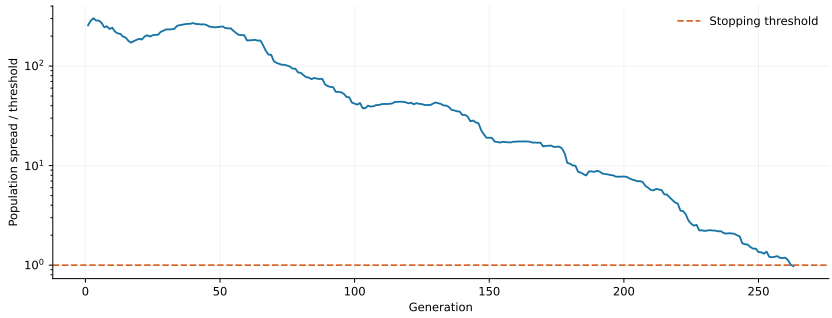
Large discharge errors receive greater weight because residuals are squared.

DE explores candidates and retains improvements



Each candidate is a parameter vector: run HBV, calculate $J = 1 - \mathrm{NSE}$, then compare.
Grey: every attempted candidate. Blue: best value so far; **61 improvements** after the first.
Trials may be worse while the running best can only decrease or stay constant.

Why did the optimization stop?

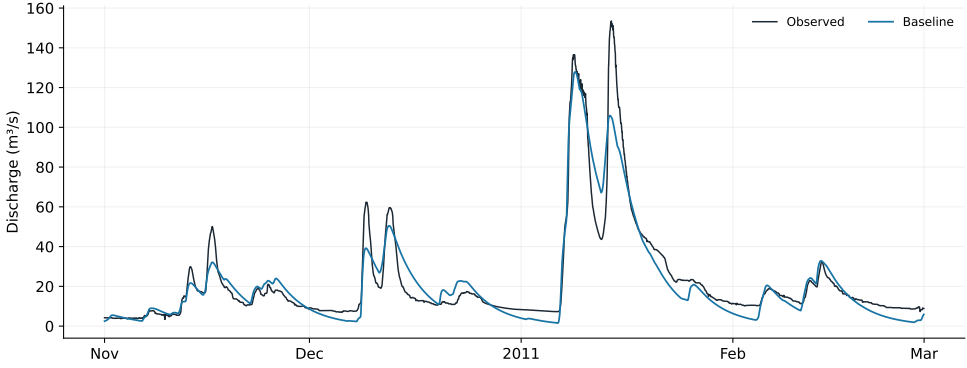


Stop when $\text{std}(J_{population}) \leq 0.001\,|\,\text{mean}(J_{population})|$.

At generation **263**: $9.10 \times 10^{-5} \leq 9.32 \times 10^{-5}$; 38,016 evaluations.

Final $J = 0.0930$ (NSE $= 0.907$). Convergence **does not prove** a global optimum or unique parameters.

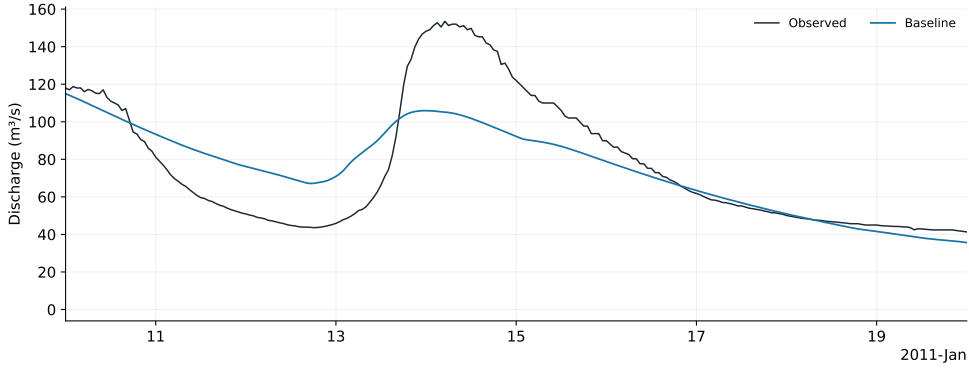
Baseline: good overall fit with an underestimated peak



NSE 0.907 — **RMSE 7.86 m^3/s** — Volume bias −5.0%

These are calibration-period metrics, not independent validation.

Baseline: zooming in on the largest observed event



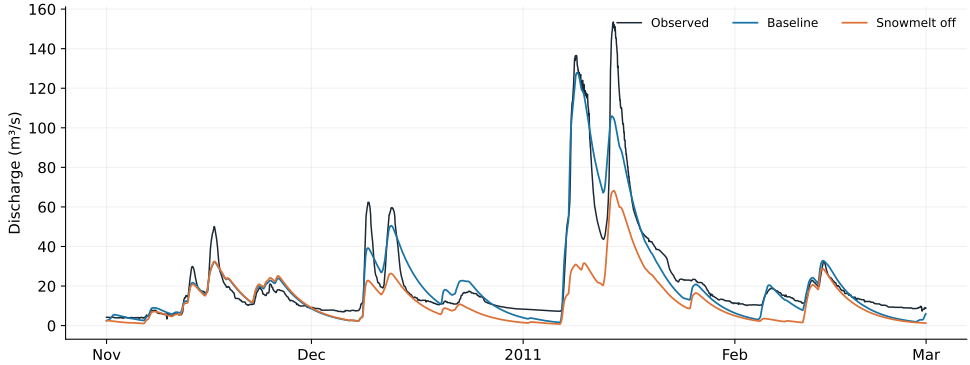
At the observed peak time: **153.5 observed vs 105.1 m^3/s simulated**.
This comparison is at the same timestamp; the simulated maximum can occur at another time.

Three process-switching experiments

Experiment	Pathways disabled	Parameter(s) set to 0
Snowmelt	Rain-induced and air-induced melt	`snw_pmf`, `snw_amf`
Channel outlets	Selected upper and lower reservoir outflows to channel	`urr_ndr`, `lrr_dre`
Subsurface	Upper and lower subsurface exports	`urr_uct`, `lrr_lct`

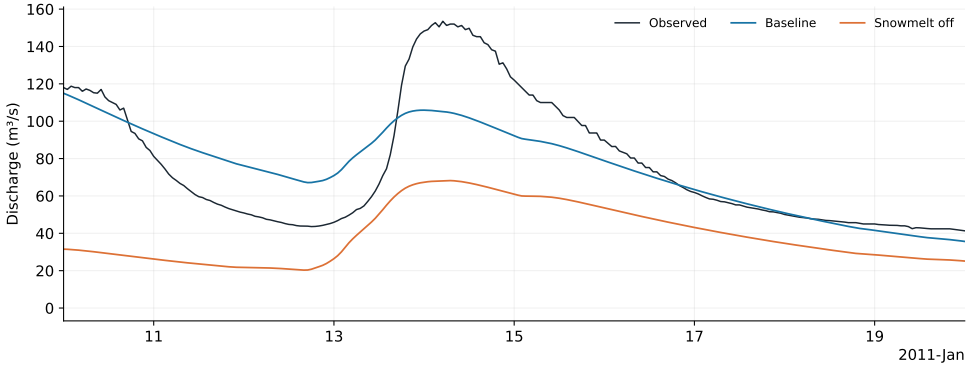
- Change one group at a time; keep other parameters at baseline values.
- Same forcing, initial states and evaluation period; **no recalibration**.

Experiment 1: turning off both snowmelt pathways



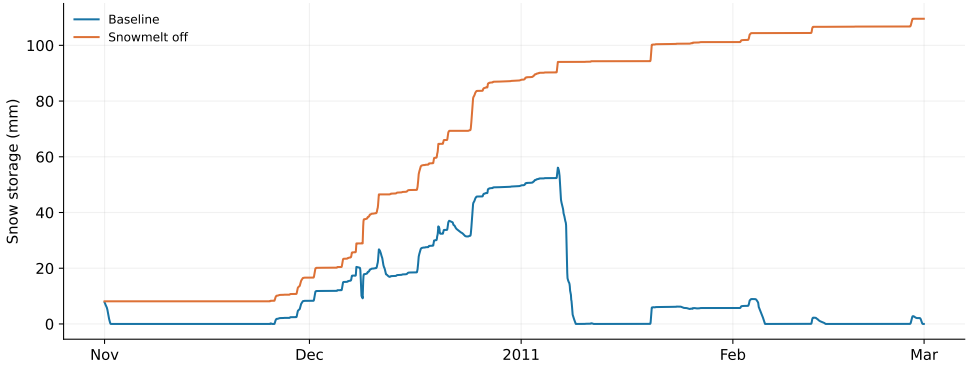
NSE **0.907** → **0.414**; volume bias −5.0% → −43.0%.
Several events are reduced when snow cannot melt through these paths.

Snowmelt: what happens around the January peak?



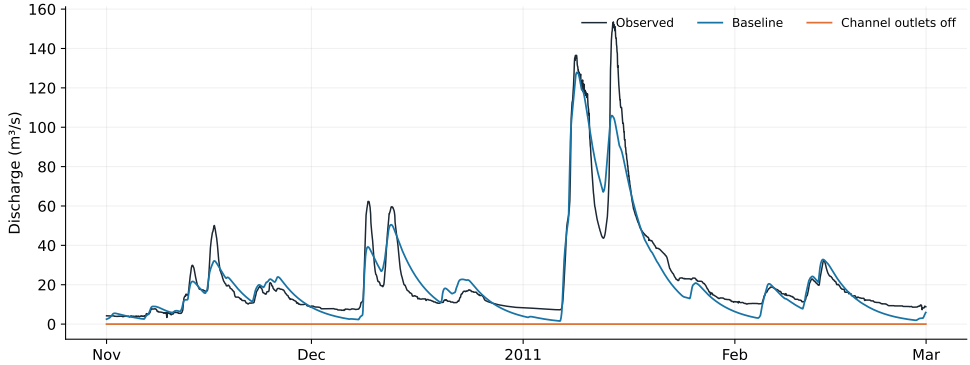
At the observed peak time: baseline 105.1; melt off 68.1; observed 153.5 m^3/s.
The change is a model response under the fixed baseline parameters.

Snowmelt: water remains in snow storage



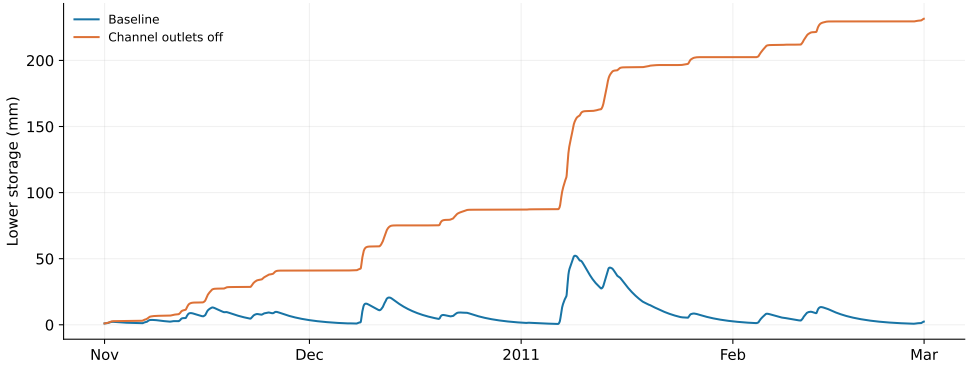
Snow can still accumulate, but the two melt terms are zero.
End-of-period snow: **109.5 mm**, compared with 0.0 mm for the baseline.

Experiment 2: selected channel outlets off



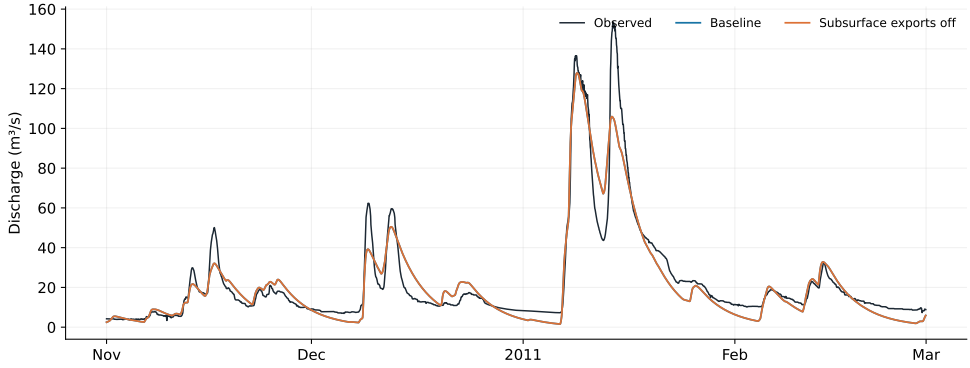
Simulated channel discharge is zero throughout this run; NSE $= -0.754$.
The separate upper threshold outlet remains enabled, but does not activate here.

Channel outlets: the lower reservoir fills



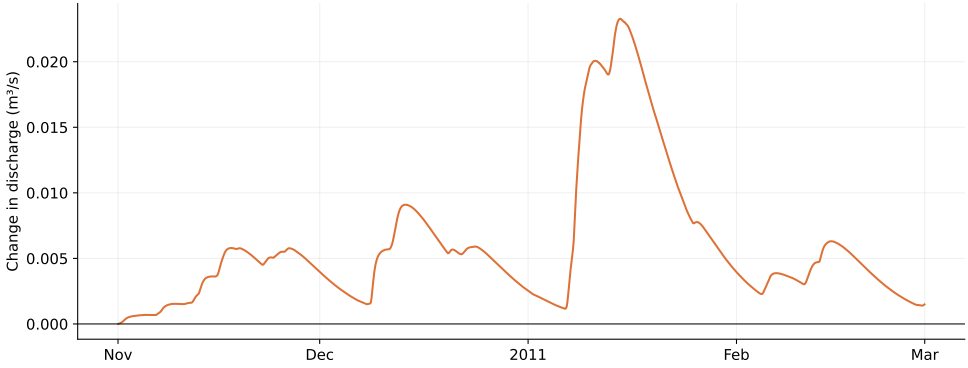
Final lower storage: **231.5 mm**, versus 2.4 mm in the baseline.
Water remains in the model rather than exiting through the disabled paths.

Experiment 3: subsurface exports off



Curves nearly overlap: NSE 0.906972 → 0.906997.
The baseline export coefficients are already close to zero.

Subsurface exports: reveal the small difference



Maximum absolute change in discharge is approximately **0.0233 $\mathbf{m}^3/\mathbf{s}$**.
Subsurface exports affect stored water and can influence channel flow indirectly.

Comparing the three experiments

Case	NSE	RMSE (m^3/s)	Volume bias
Baseline	0.907	7.86	−5.0%
Snowmelt off	0.414	19.73	−43.0%
Selected channel outlets off	−0.754	34.12	−100.0%
Subsurface exports off	0.907	7.86	−5.0%

- The first two interventions have large effects on discharge and storage.
- The third has little effect under this calibrated parameter set.
- NSE can be negative; zero corresponds to predicting the observed mean.

Interpretation: what do these experiments tell us?

- **Snowmelt:** freezing melt terms holds water in snow and changes the timing and volume of runoff.
- **Channel outlets:** disabling two outflows disrupts the discharge pathway and increases storage.
- **Subsurface:** a near-zero baseline coefficient leaves little immediate effect to remove.

These are model-specific responses, not direct measurements of process contribution in the catchment.

Limitations and next steps

- No recalibration after switching: parameter compensation has not been tested.
- The full record is used for calibration, with no warm-up or held-out validation.
- NSE emphasizes squared errors; also assess peak timing, volume and low flows.
- One DE run reached its stopping criterion, which does not establish a unique or global optimum.

Next: Recalibrate selected simplified models and evaluate them on independent data.

Conclusions and questions

1. Baseline NSE is 0.907, but the largest observed peak is underestimated at its timestamp.
2. Turning off melt or selected channel outlets substantially changes discharge and storage.
3. Turning off subsurface exports barely changes this particular baseline simulation.

Discussion: Which process experiments should we recalibrate? What validation period and performance metrics should we use?